NSSC 2026 — Data Analytics

Unsupervised Anomaly Detection in Mars HiRISE Imagery

Convolutional Autoencoder · Isolation Forest · Reconstruction Heatmaps

Key results. From-scratch ConvAE → Isolation Forest on latents → MAD threshold → post-threshold heatmaps. IsolationForest `contamination='auto'` is recorded but **not** used to set the anomaly count.

$$d_{\text{lat}} = 512, \quad \tau = 0.5253, \quad N_{\text{flag}} = 242/10422$$

$$\mathcal{L} = 0.40\,\text{MSE} + 0.35\,(1 - \text{SSIM}) + 0.25\,\|\nabla x - \nabla \hat{x}\|_1$$

1. Problem setup

HiRISE on MRO produces a large archive of Martian surface crops. In this challenge the released set (~10k grayscale 227×227 frames) has no labels, while a hidden mix of sensor artifacts, spliced terrain, and non-Martian content was injected at unknown rate. The task is unsupervised novelty detection.

2. Phase 1 — Latent compression

2.1 Architecture

We train a convolutional autoencoder (ConvAE) entirely from scratch — no pretrained backbones. Each crop maps to a fixed-length latent for Isolation Forest.

$$z \in \mathbb{R}^{512}$$

Encoder: pad 227→228, four stride-2 blocks to 14×14×256, squeeze to 128, FC bottleneck. Decoder: nearest upsample + convs, final resize to 227. Latent dimension **512**.

2.2 Objective

MSE alone over-smooths ridgelines, so we add SSIM and a finite-difference gradient penalty (no VGG/ResNet perceptual nets).

$$\mathcal{L} = 0.40\,\text{MSE} + 0.35\,(1 - \text{SSIM}) + 0.25\,\|\nabla x - \nabla \hat{x}\|_1$$

Final training objective (after curriculum).

Curriculum: MSE warmup → add SSIM/grad → raise weights → fine-tune edges. Jumps on the composite curve are expected when new terms turn on — not divergence. Optimiser: AdamW + cosine + grad clip.

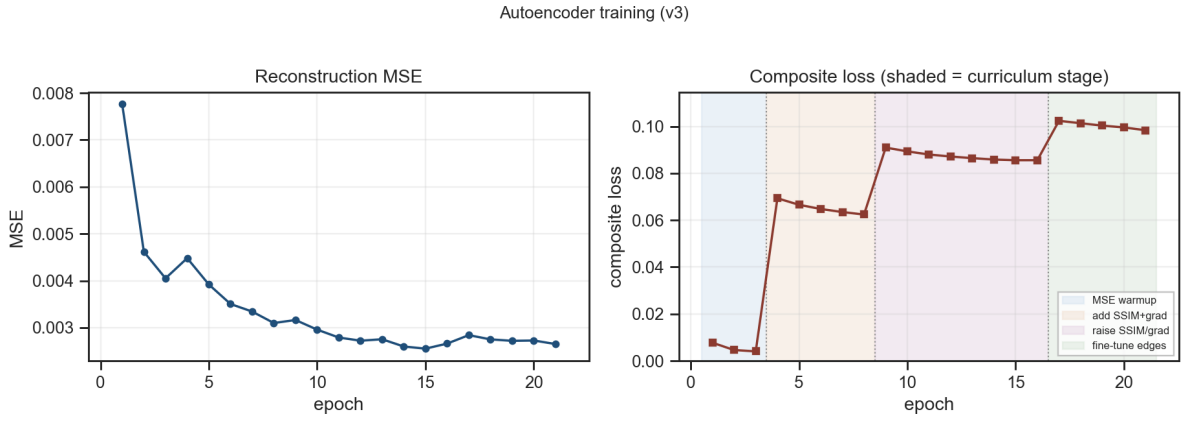


Fig. 1 — Left: reconstruction MSE. Right: composite loss with shaded curriculum stages (legend); vertical jumps mark loss-term changes.

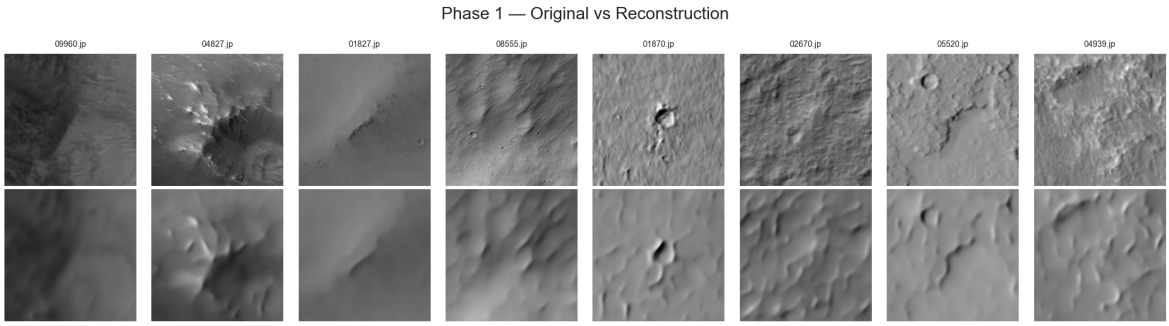


Fig. 2 — Originals (top) vs reconstructions (bottom).

2.3 Latent diagnostics

t-SNE (perplexity 30, 3000-point subsample) is a qualitative check. Colour = novelty, not class labels.

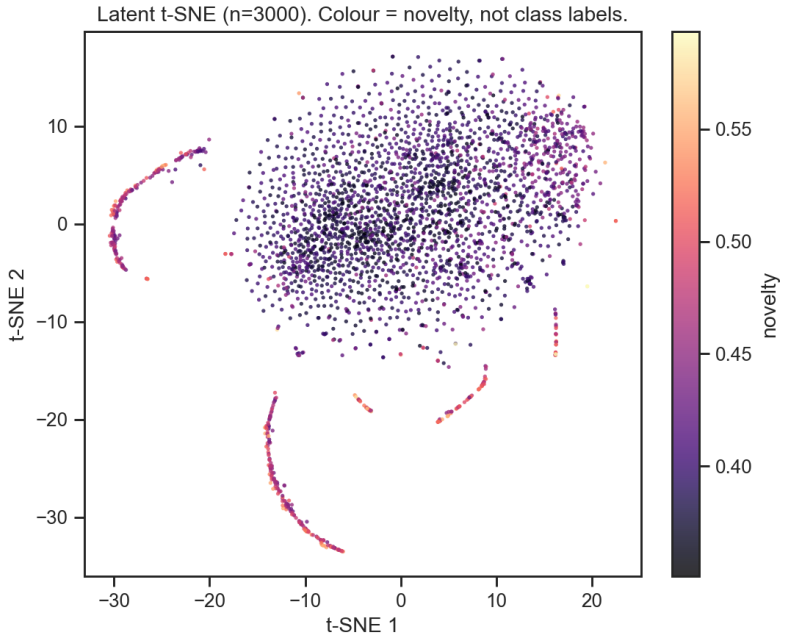


Fig. 3 — Low-novelty core; higher-novelty points toward the fringe.

3. Phase 2 — Novelty scoring

3.1 Isolation Forest on latents

Isolation Forest (400 trees) on standardised 512-D latents. Sklearn `score_samples` is higher for inliers; we invert it:

$$s = -\text{score_samples}(\tilde{z}), \quad \tilde{z} = \text{StandardScaler}(z), \quad (\uparrow s = \text{more anomalous})$$

3.2 Threshold

No fixed top-N. Elbow (rank 517, 0.503), Gaussian reference, and MAD / modified-z with k=3.5.

$$\tau_{\text{G}} = \mu_s + 2.5\,\sigma_s$$

$$\tau_{\text{MAD}} = \text{median}(s) + \frac{3.5}{0.6745}\,\text{MAD}(s)$$

Selected: MAD τ=0.5253, **242** crops. `contamination='auto'` does not set the count.

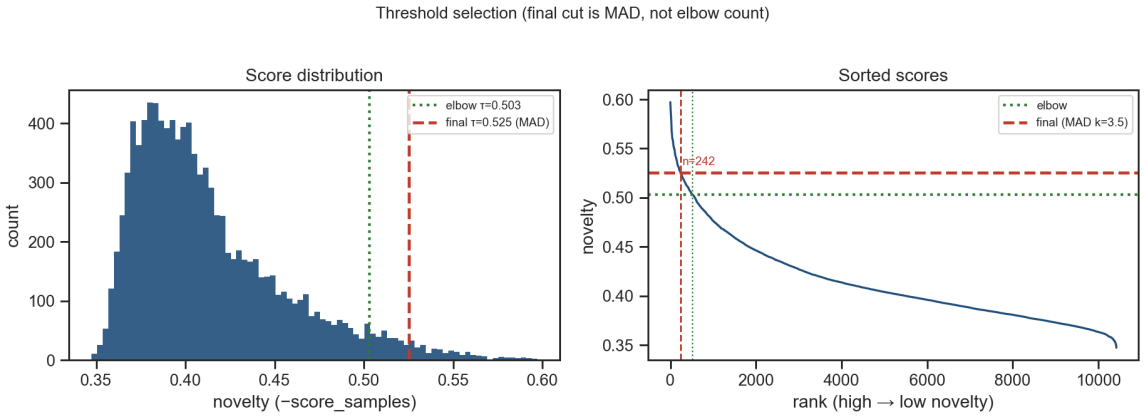


Fig. 4 — Green: elbow. Red: final MAD threshold.

3.3 Source metadata

Join via source_image_id. Released longitude is only 0°/180°, so we plot lat / sun / season and a jittered coordinate panel.

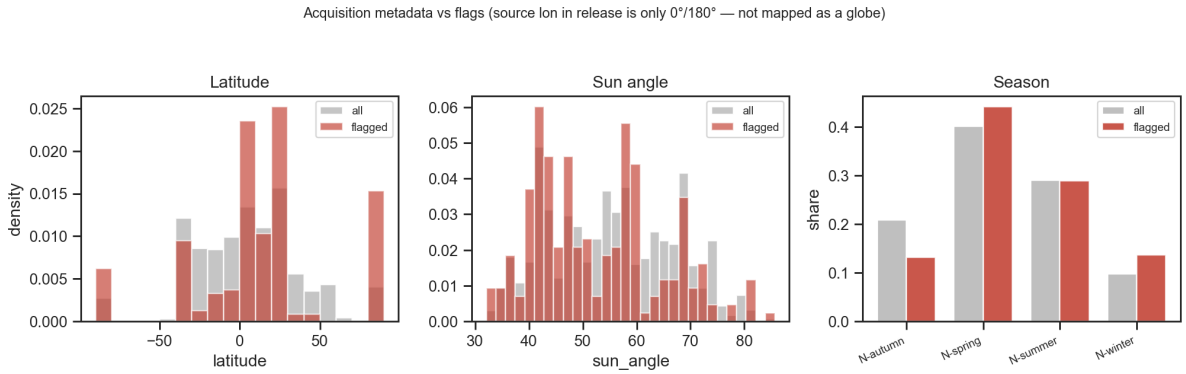


Fig. 5 — Acquisition covariates.

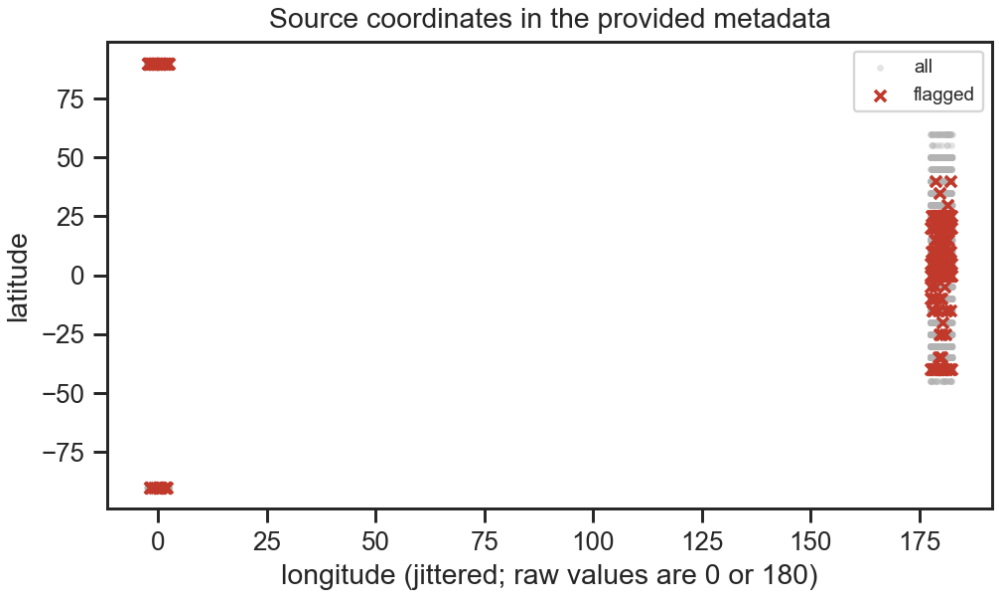


Fig. 5b — Source coordinates (lon jittered).

3.4 Optional metadata fusion

Concatenate standardised latent with acquisition features; refit IF:

$$u = [\, z;\ \tilde{\theta}_{\text{sun}};\ \tilde{\phi}_{\text{lat}};\ \tilde{\lambda}_{\text{lon}};\ \tilde{r};\ \mathbf{e}_{\text{season}} \,]$$

Tildes = hand-normalised covariates; then StandardScaler on both blocks. Fused cut μ+2.5σ = 0.5206. Image-only 242; fused 298; overlap 221 (exclusive 21/77).

Pipeline	Features	Cutoff	Flags
Primary	512-D latent	MAD · τ=0.5253	242
Fusion	latent + acquisition	μ+2.5σ · τ=0.5206	298
Agreement	both	—	221

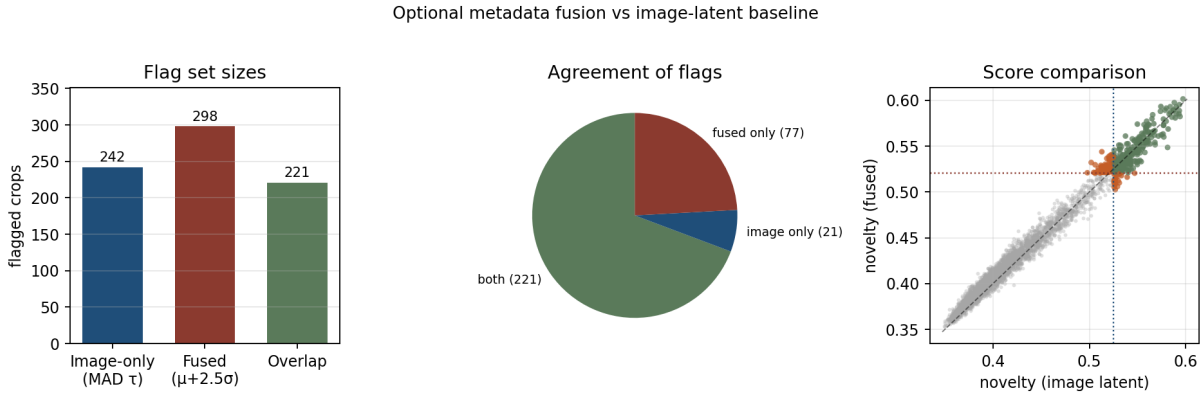


Fig. 7 — Fusion vs image-only.

Decision. Official answer = image-latent MAD pipeline.

4. Phase 3 — Reconstruction interpretability

Top five among novelty ≥ τ (post-threshold). Pixel error:

$$E_{ij} = |x_{ij} - \hat{x}_{ij}|$$

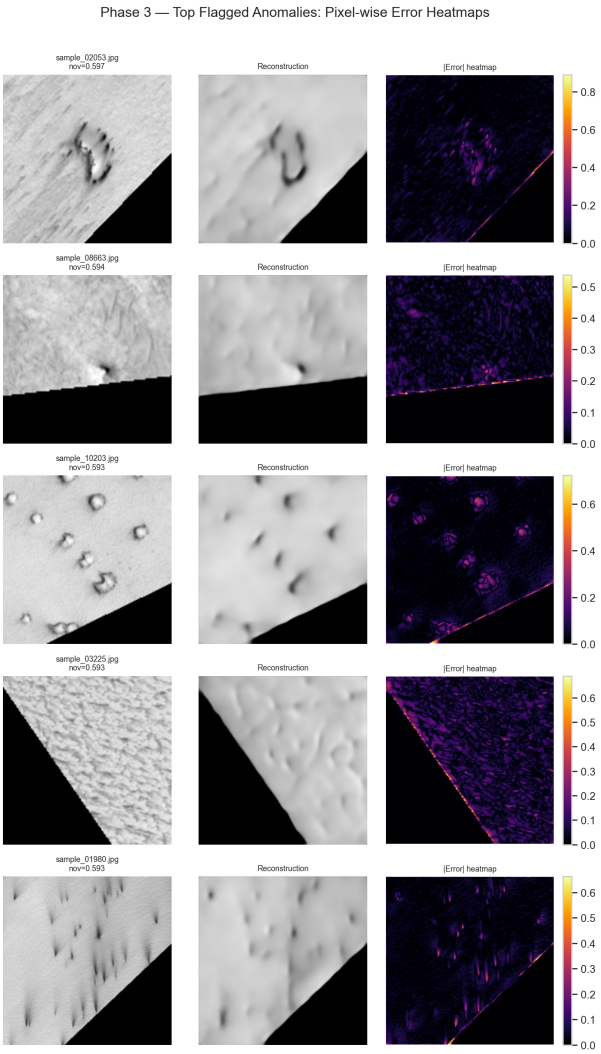


Fig. 6 — Top-5: input, reconstruction, |error|.

4.1 Working hypotheses

Hypotheses for discussion, not claimed ground truth.

`sample_02053.jpg` (novelty=0.597, SRC_064): Linear dark band / hard crop edge — sensor streak or framing mask.

`sample_08663.jpg` (novelty=0.594, SRC_097): Diffuse blotchy texture — possible domain shift / atypical content.

`sample_10203.jpg` (novelty=0.593, SRC_166): Dense ripple field — rare dune fabric or high-frequency splice.

`sample_03225.jpg` (novelty=0.593, SRC_039): Compact landform with rim error — candidate seam between patches.

`sample_01980.jpg` (novelty=0.593, SRC_044): Speckled bumps / patchy residuals — local corruption or pasted fragments.

5. Phase 4 — Design iterations

Symptom → diagnosis → fix in `MODEL_CHANGELOG.md`.

6. Summary of answers

Item	Answer
Latent dimension	**512**
Loss (final)	see §2.2 equation

Novelty definition	see §3.1 equation
IF contamination	`auto`
Threshold τ	**0.525258** (MAD, k=3.5)
Flagged count	**242 / 10422**
Top-5 (post-τ)	sample_02053.jpg, sample_08663.jpg, sample_10203.jpg, sample_03225.jpg, sample_01980.jpg

$$\mathcal{L} = 0.40\,\mathrm{MSE} + 0.35\,(1 - \mathrm{SSIM}) + 0.25\,\|\nabla x - \nabla \hat{x}\|_1$$

$$s = -\mathrm{score_samples}(\tilde{z}), \quad \tilde{z} = \mathrm{StandardScaler}(z), \quad (\uparrow s = \text{more anomalous})$$

$$\tau_{\mathrm{MAD}} = \mathrm{median}(s) + \frac{3.5}{0.6745}\,\mathrm{MAD}(s)$$

Accompanying: notebook, MODEL_CHANGELOG.md, submission/.